

Advaith Vecham

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EDUCATION

University of Washington, Seattle

Sep 2025 – Jun 2029

Bachelor of Science in Computer Science

Relevant Coursework: Operating Systems, Algorithms, Machine Learning, Object-Oriented Programming, Advanced Calculus

EXPERIENCE

Linux Kernel Contributor - Rust for Linux Subsystem

Aug 2026 – Present

- Added safe `Regulator` methods in `rust/kernel/regulator.rs` — `get/set_current_limit()`, `get/set_mode()` with a type-safe `Mode` enum, and `set_load()` — over the C `regulator_*` API.
- Wrote the `unsafe` FFI bindings with `# Safety` invariants, `Result`-based `errno` handling, and `rustdoc/doctests`.
- Built under `CONFIG_RUST` and submitted to LKML (DCO-signed, via `git send-email/b4`) through maintainer review.

PROJECTS

SemiLLM: Layer-Streaming Mixed-Precision Inference Engine

C++, CUDA, Python

- Built an execution engine that runs LLMs larger than VRAM by streaming quantized transformer layers CPU→GPU over an overlapped CUDA copy/compute pipeline.
- Chose each layer's precision dynamically from available bandwidth and layer sensitivity, spending more bits only where accuracy needs them.
- Ran faster than uniform-quantization streaming and the AirLLM baseline at the same accuracy.

Insuo: Autonomous Browser Agent

Python, TypeScript, C++, WebKit, Tauri, Gemini API

- Architected a hierarchical plan-first agent (Goal → Parts → Steps) that decomposes tasks into verifiable sub-goals, cutting LLM calls ~60% vs. pure ReAct by executing deterministic steps open-loop.
- Designed self-healing loops that detect failed steps and reset the agent to retry with improved context; added human-in-the-loop pause/approve/stop controls.
- Implemented adaptive context windowing with model-aware token budgets and turn-level truncation to fit multimodal transcripts within provider limits (8K–1M tokens); packaged as a native desktop app via Tauri.

Weight-Switching CoT: Multicultural LoRA Routing for Small Models

Python, PyTorch

- Treated a language shift in a model's own chain-of-thought as a routing signal, hot-swapping language/culture-specific LoRA experts *mid-generation* over a single continuous KV cache.
- Swapped only LoRA deltas (not full models) so the shared cache stays valid, letting a small model reason natively across languages without loading separate full-size models.
- Ran controlled experiments isolating the benefit of the experts from the overhead of sharing one cache.

CHIP-8/Superchip Emulator

Rust

- Engineered a cycle-accurate CHIP-8 interpreter: 4KB memory, 16 registers, hardware stack, and a fetch-decode-execute loop parsing all 35 16-bit opcodes via bitwise ops.
- Rendered the 64x32 monochrome display at 60Hz with XOR-based sprite collision detection, and built a CLI debugger (register dumps, memory stepping) that passes the Corax+ test ROM.

Simbiote: Offline Scan-to-Sim Robot Training Platform

Jul 2026

1st Place, Dell x NVIDIA Hackathon (Seattle): Dell Pro Max GB10 (~\$6,500)

OpenUSD, MuJoCo, RL

- Directed a 4-person team building an air-gapped *scan* → *simulate* → *train* → *operate* platform running entirely offline on the Dell Pro Max (GB10) behind a local Nemotron model.
- An iPhone LiDAR scan becomes a sim-ready OpenUSD scene; a mobile manipulator is trained with RL for collision-free navigation and grasping, then driven by hand-tracking teleoperation or plain-language commands — with every session logged as a demonstration that fine-tunes the policy (BC → PPO), closing a sim-to-real-to-sim loop.

STRUCT: Text-to-Robot Rapid Prototyping Pipeline

Aug 2026

NVIDIA Dev Champions (2nd), NVIDIA Sparkhacks (Seattle)

DGX Spark, LangGraph, Isaac Sim, Kaolin

- Built an end-to-end pipeline that rapidly prototypes robots from a prompt — CAD design, PCB design, teleoperation-data generation for model training, and RL — all coded and orchestrated by a local Nemotron-Lightning / Laguna-S 2.1 model quantized to 3-bit on a single DGX Spark (GB10), under an “agent proposes, deterministic harness disposes” contract.
- Owned the 3D reconstruction + RL training stage: from a single video/image, generated 8 sim-ready “digital cousin” scene variants using NVIDIA Kaolin and Omniverse NuRec neural reconstruction for Isaac Sim, with VoMP volumetric material properties.

HONORS & AWARDS

- Y Combinator Startup School — Selected (5,000 of 35,000+ applicants).** Jul 2026
Attended talks by Jensen Huang (NVIDIA), Sam Altman (OpenAI), and Alexandr Wang (Meta/Scale AI); pitched Insuo to YC partner Jared Friedman; received startup credits and resources.
- IMC Prosperity Challenge 4 — Top 10%.** Implemented mean-reversion and market-making strategies. Apr 2026
- USACO Gold.** 30,000 participants; top 10% qualify. Jan 2025
- AIME Qualifier (x2).** Consistently scoring 90+. Nov 2022

SKILLS

Languages: Rust, Python, C/C++, CUDA, JavaScript/TypeScript, SQL, Swift

Frameworks & Libraries: PyTorch, TensorRT-LLM, LangGraph, MuJoCo, Isaac Sim, Kaolin, OpenUSD, Playwright, SDL2, WebKit

Tools & Platforms: Linux, Git/GitHub, Docker, NVIDIA DGX Spark (GB10), AWS, SQLite, macOS Keychain, CI/CD

Systems & ML Concepts: Operating Systems, Concurrency, Memory/Cache Optimization, Quantization (AWQ/GPTQ), LLM Inference, Agent Architectures, Reinforcement Learning, Sim-to-Real